11 A balloon is being inflated. The balloon is modelled as a sphere with radius x cm at time t s. The volume $V\,\text{cm}^3$ is given by $V = \frac{4}{3}\pi x^3$.

The rate of increase of volume is inversely proportional to the radius of the balloon. Initially, when $t = 0$, the radius of the balloon is 5 cm and the volume of the balloon is increasing at a rate of $21\,\text{cm}^3\,\text{s}^{-1}$.

(a) Show that x satisfies the differential equation $\frac{\mathrm{d}x}{\mathrm{d}t} = \frac{105}{4\pi x^3}$. **[5]**

(b) Find the radius of the balloon after two minutes. **[5]**

(c) Explain why the model may not be suitable for very large values of t. **[1]**

11(a)

11(b)	
11(c)	